

Unit I Basics of Machine Learning

Unit I – Basics of Machine Learning

1. Machine Learning Kya Hai?

Machine Learning ek aisi technique hai jisme computer data ko dekh kar patterns seekhta hai aur future predictions ya decisions karta hai. Isme har rule ko manually likhne ki jagah machine ko data se seekhne ka mauka diya jata hai.

Important Characteristics

- ? Data-driven approach
- ? Pattern recognition
- ? Predictive capability
- ? Ability to improve with more data

Main Goals

- ? Prediction
- ? Classification
- ? Clustering
- ? Pattern discovery

2. Machine Learning ke Types

Supervised Learning

- ? Labeled data par train hoti hai.
- ? Input aur output dono known hote hain.
- ? Examples: regression, classification.

Unsupervised Learning

- ? Labeled data ke bina kaam karti hai.
- ? Data ko groups ya clusters mein organize karti hai.
- ? Examples: clustering, PCA.

Reinforcement Learning

- ? Agent environment ke saath interact karta hai.
- ? Reward ke basis par seekhta hai.
- ? Examples: robotics, game playing.

3. Scope of Machine Learning

Machine Learning ka use healthcare, finance, banking, e-commerce, social media, recommendation systems, NLP, computer vision, robotics aur automation mein hota hai. Iski demand isliye badh rahi hai kyunki bohot saare real-world problems ko data se solve kiya ja sakta hai.

4. Limitations of Machine Learning

Machine Learning ke kaam karne ke liye quality data chahiye hota hai. Agar data incomplete, biased ya noisy hai to output bhi galat ho sakta hai. Kuch major limitations hain:

- ? Data dependency
- ? Overfitting aur underfitting
- ? Interpretability ka issue
- ? High computational cost
- ? Need for large datasets

5. Regression

Regression supervised learning ka ek important part hai jisme continuous output predict kiya jata hai. Examples:

- ? House price prediction
- ? Sales forecasting
- ? Temperature prediction

Common Regression Types

- ? Linear Regression
- ? Polynomial Regression
- ? Ridge/Lasso Regression

Key Idea

? Independent variables aur dependent variable ke beech relationship establish ki jati hai.

6. Probability and Statistics

Probability events ki likelihood ko describe karta hai. Statistics data ko collect, analyze aur interpret karne ke liye use hoti hai.

Important Concepts

? Mean

? Median

? Mode

? Variance

? Standard deviation

? Covariance

? Correlation

Why These Matter

Ye concepts data ko samajhne aur model performance evaluate karne mein help karte hain.

7. Linear Algebra for Machine Learning

Linear algebra machine learning ka mathematical foundation hai.

Important Topics

? Vectors

? Matrices

? Dot product

? Eigenvalues

? Eigenvectors

Why Important

Neural networks, optimization aur dimensionality reduction mein linear algebra ka bahut use hota hai.

8. Convex Optimization

Optimization ka matlab hota hai best solution dhoondhna. Machine learning mein optimization ka use loss function ko minimize karne ke liye hota hai.

Key Idea

? Model ke parameters ko update karna taki error kam ho.

Common Methods

? Gradient descent

? Stochastic gradient descent

? Convex optimization techniques

9. Data Preprocessing

Data preprocessing machine learning pipeline ka important part hai. Isme data ko training-ready banaya jata hai.

Main Steps

? Missing values handle karna

? Outliers detect karna

? Categorical values ko numeric mein convert karna

? Feature scaling

? Train-test split

Why Important

Achi preprocessing se model better aur reliable ban sakta hai.

10. Data Visualization

Data visualization se data ko visually samjha ja sakta hai.

Common Visuals

- ? Scatter plot
- ? Histogram
- ? Bar chart
- ? Box plot
- ? Heatmap

Usefulness

Ye trends, outliers aur distribution ko identify karne mein help karta hai.

11. Hypothesis Function and Hypothesis Testing

Hypothesis Function

Hypothesis function model ka proposed relationship hota hai jo input ko output se connect karta hai.

Hypothesis Testing

Ye check karta hai ki observed result statistically significant hai ya nahi.

Important Terms

- ? Null hypothesis
- ? Alternate hypothesis
- ? Significance level

12. Overfitting, Underfitting and Bias-Variance Tradeoff

Overfitting

Model training data ko bahut zyada achhe se yaad kar leta hai, lekin new data par weak hota hai.

Underfitting

Model data ke patterns ko properly samajh nahi pata.

Bias-Variance Tradeoff

- ? High bias => underfitting
- ? High variance => overfitting

13. Important Exam Points

- ? Machine Learning data se seekh kar predictions karti hai.
- ? Supervised learning mein labels hote hain.
- ? Unsupervised learning mein labels nahi hote.
- ? Regression continuous output ke liye use hoti hai.
- ? Probability aur statistics model evaluation ke liye important hain.
- ? Data preprocessing model accuracy ko improve karta hai.
- ? Overfitting aur underfitting ko samajhna very important hai.